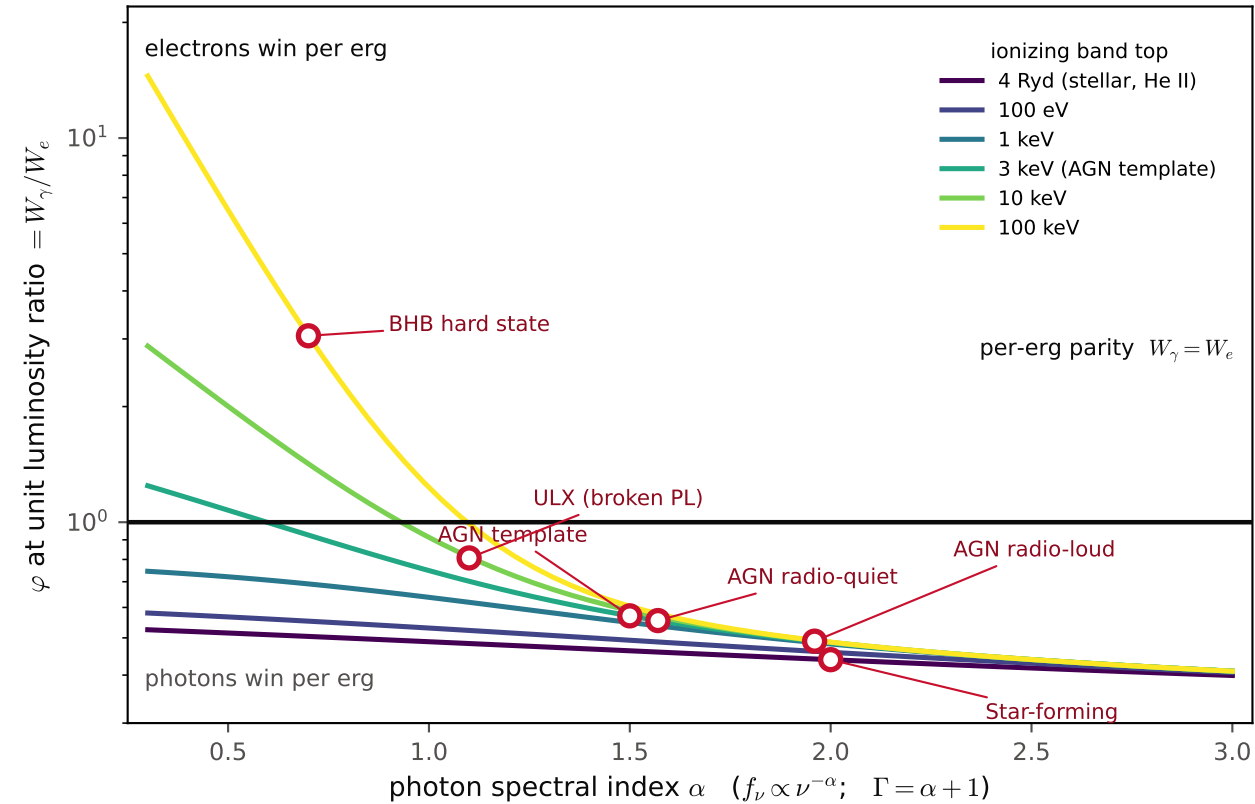
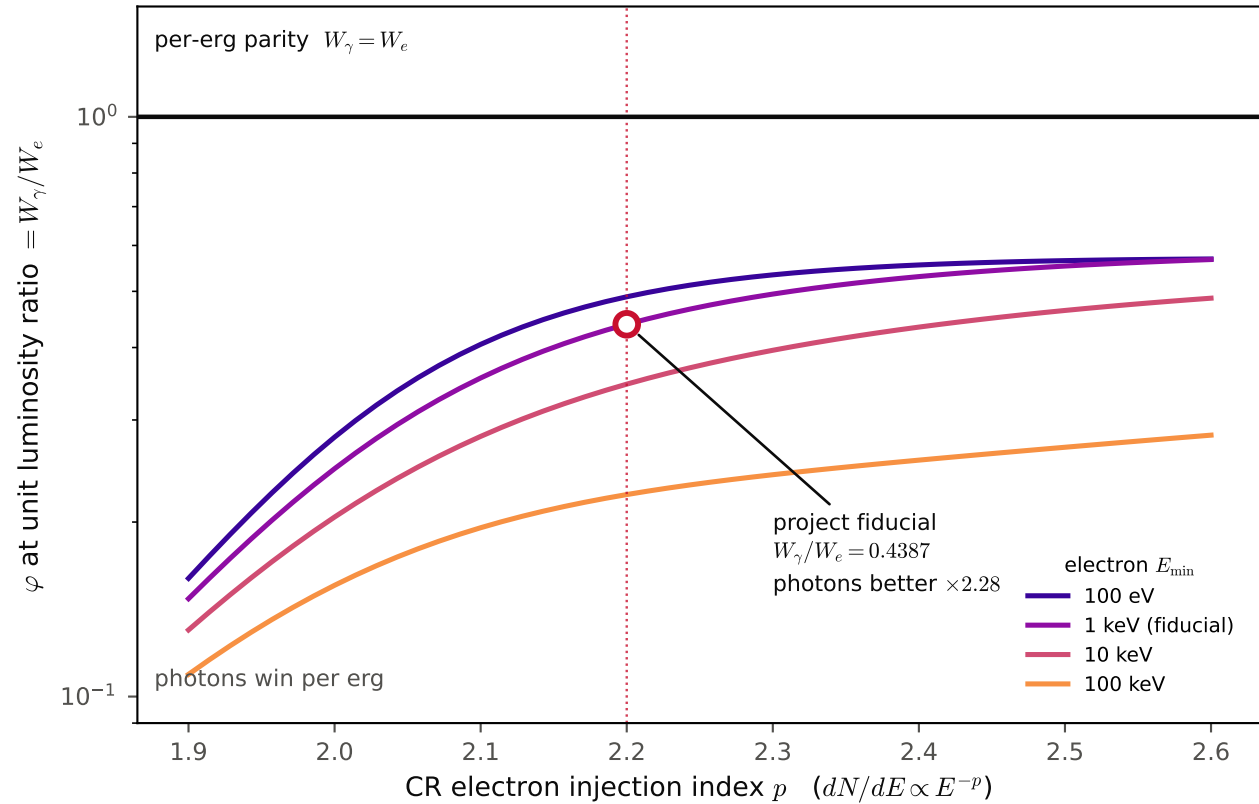


(a) the photon spectrum, at fixed CR electrons



(b) the CR electron spectrum, at fixed photons



IGM at $z=10$, $x_e=10^{-4}$, pure H. $W \equiv \langle E \rangle / \langle N \rangle$: the energy per ionization ACTUALLY ACHIEVED, so $\varphi > 1$ means CR electrons ionize more per erg than photons do.

Photon yields carry the IGM transparency factor $1 - e^{-\tau(E)}$ over one Hubble length: $\tau=1.93$ at 1 keV, 0.048 at 3 keV — the $z=10$ IGM is thick below ~ 500 eV and transparent above ~ 3 keV, so a hard source's photons are lost while its electrons deposit.

Electron yields: the loss+IC route (igm_losses.py). $f_{\text{dep}}=1$ (assumption A11): every injected electron is assumed to deposit, so the electron channel is a CEILING, not an estimate.

Class markers sit at each class's OWN index and band top — stellar Marques-Chaves+26; AGN Telfer+02 and Graziani+18; ULX and BHB Gladstone+09 (the BHB index is second-hand, status C). Curves are exact, not sampled: ζ is linear in the emissivity, which check S2 tests rather than assumes.